Week 5 Assignment 2: Configuring Site-to-Site VPNs

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CS-CNS09-25030

Contents

Introduction

I engaged in configuring and monitoring an IPSec VPN tunnel between two routers using Cisco IOS. The goal was to ensure that traffic between the LAN on R1 and the LAN on R3 was securely encrypted using the IPSec protocol, triggered by interesting traffic defined through access control lists. As part of the validation process, I used the show crypto ipsec sa command to observe the state of the tunnel, examine packet encryption and decryption statistics, and verify the active Security Associations (SAs). This exercise provided practical insight into how IPSec tunnels are established, maintained, and verified in a live environment.

Part 1

Enabling Security Features

To begin configuring the Site-to-Site VPN, I first needed to ensure that the security features required for VPN functionality were available on Router R1. I accessed R1 through the CLI in Cisco Packet Tracer and entered privileged EXEC mode using the enable command. To check if the required security package was active, I issued the show version command and examined the "Technology Package License Information" section. I noticed that the securityk9 module was not active, as it was listed as "None."

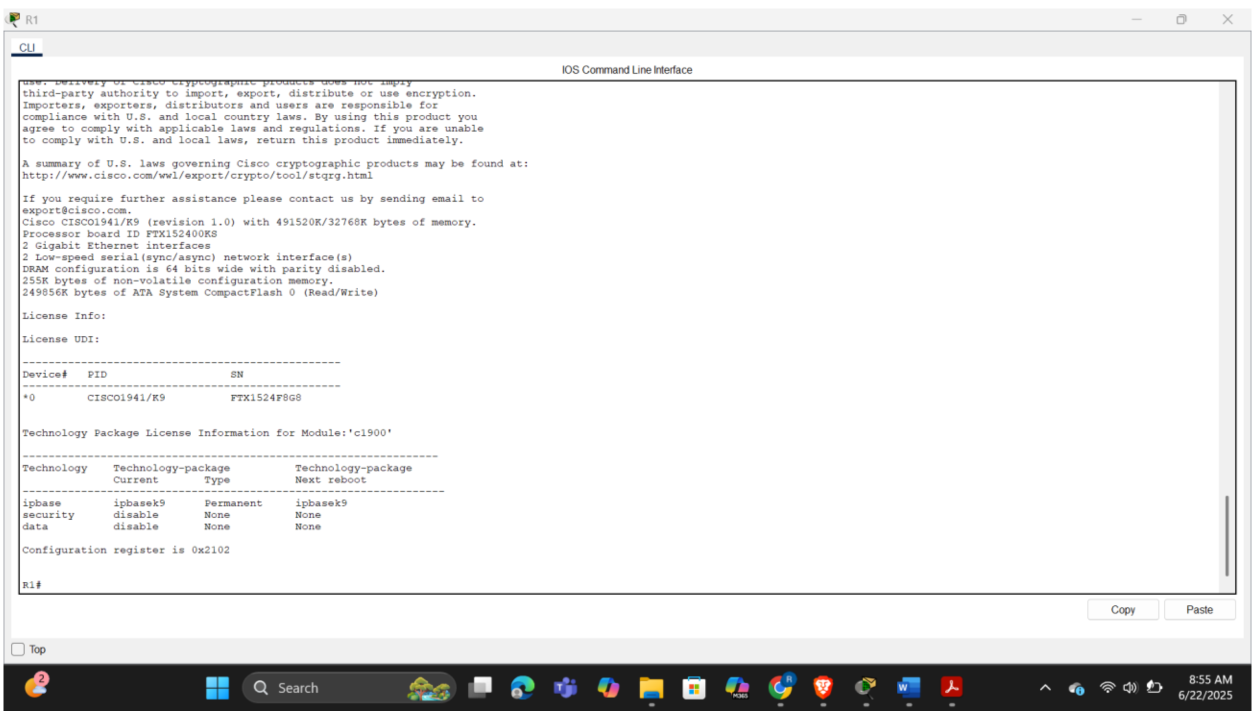

Fig 1.0 verifying security parameters on R1 using show version command

To activate it, I entered global configuration mode with configure terminal and executed the command license boot module c1900 technology-package securityk9 to instruct the router to enable the security package at the next boot. After that, I exited configuration mode by typing end and saved the current running configuration to the startup configuration using copy running-config startup-config to ensure the license change would persist after reboot.

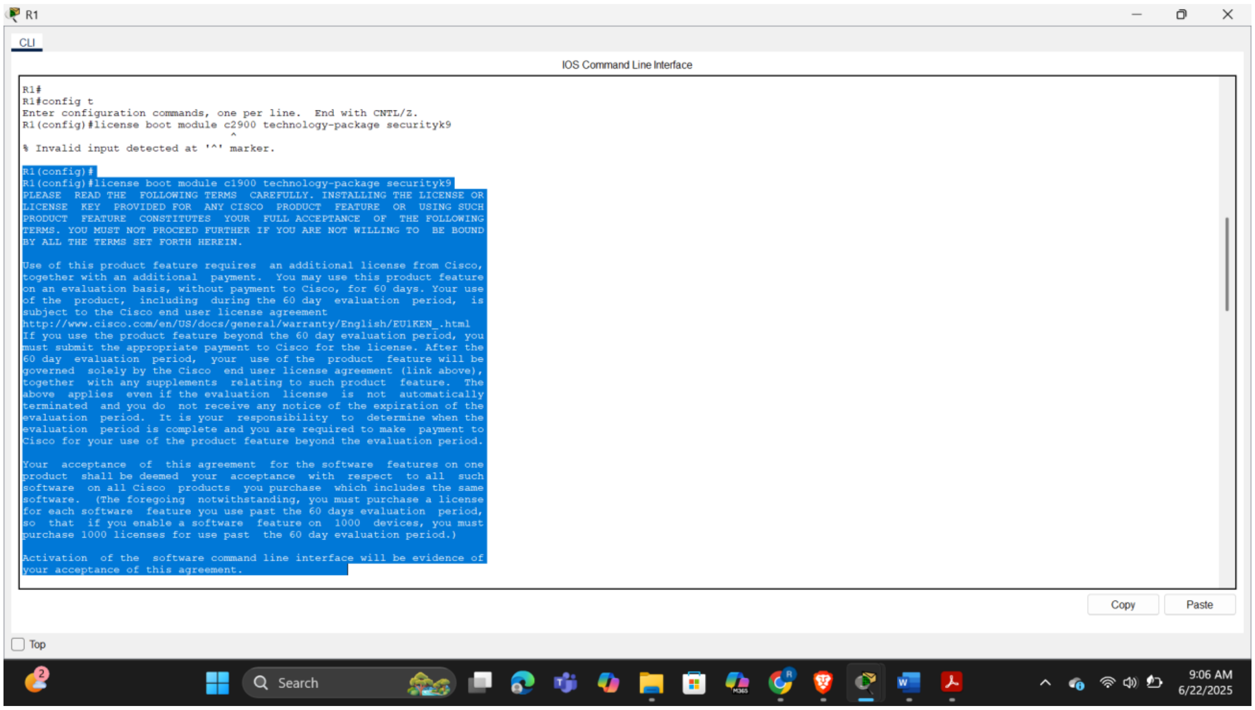

Fig 1.1 activating k9 security module on R1

I then reloaded the router with the reload command and confirmed the prompt to restart. Once the router rebooted, I again used the show version command to confirm the activation. This time, the output showed that the securityk9 technology package was in evaluation mode and successfully activated, confirming that Router R1 was now equipped with the necessary security capabilities to proceed with VPN configuration. Once the router rebooted, I verified that the security technology package had been successfully enabled by running the show version command. In the output, I looked for the “securityk9” feature and confirmed that its status was “Enabled.” This indicated that the router now had the cryptographic capabilities required to configure IPsec VPNs.

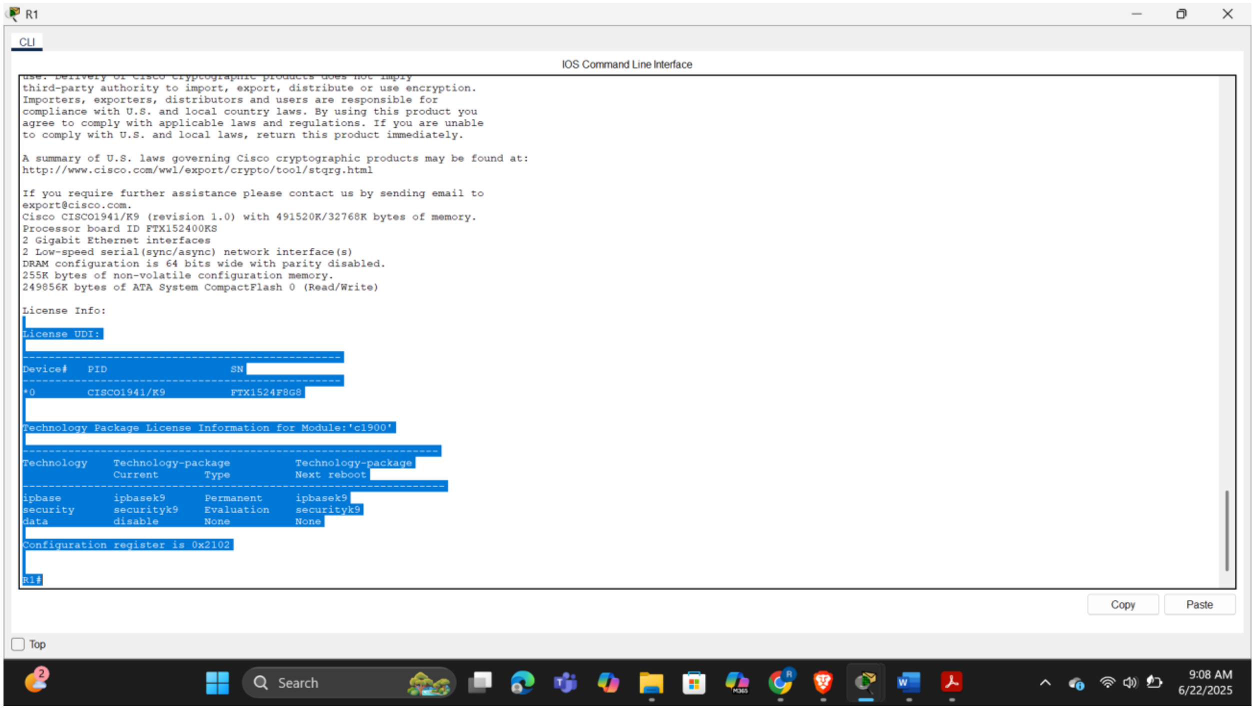

Fig 1.2 verifying the enabled K9 security on R1

Part 2

Configuring IPsec Parameters on R1

To verify baseline connectivity during the IPsec configuration process on R1, I initiated a ping from PC-A to PC-B. Since both devices reside within the same local network and are not part of the defined "interesting traffic" for the VPN, this test confirmed that local communication was functioning properly without encryption. The successful ping indicated that IP addressing, static routing, and interface configurations were correctly set up and operational. This step ensured that the network was behaving as expected prior to enabling IPsec features.

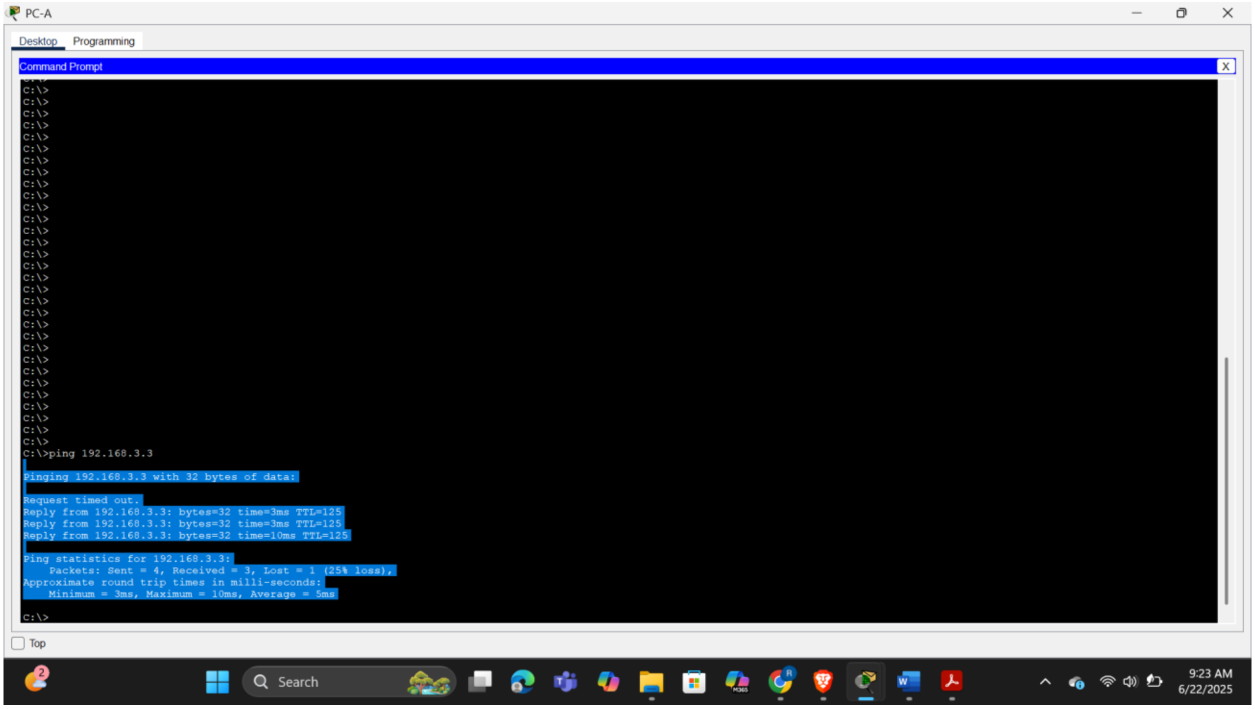

Fig 1.3 pinging pc B from pc A

Identifying Interesting Traffic

With cryptographic capabilities in place, I proceeded to define what traffic should trigger the VPN tunnel—known as "interesting traffic." This was accomplished by creating an Access Control List (ACL) using the command access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255. This ACL instructs the router to consider traffic between the LAN behind R1 (192.168.1.0/24) and the LAN behind R3 (192.168.3.0/24) as traffic that should be encrypted via IPsec. The ACL did not require an explicit deny statement because Cisco ACLs include an implicit "deny all" at the end.

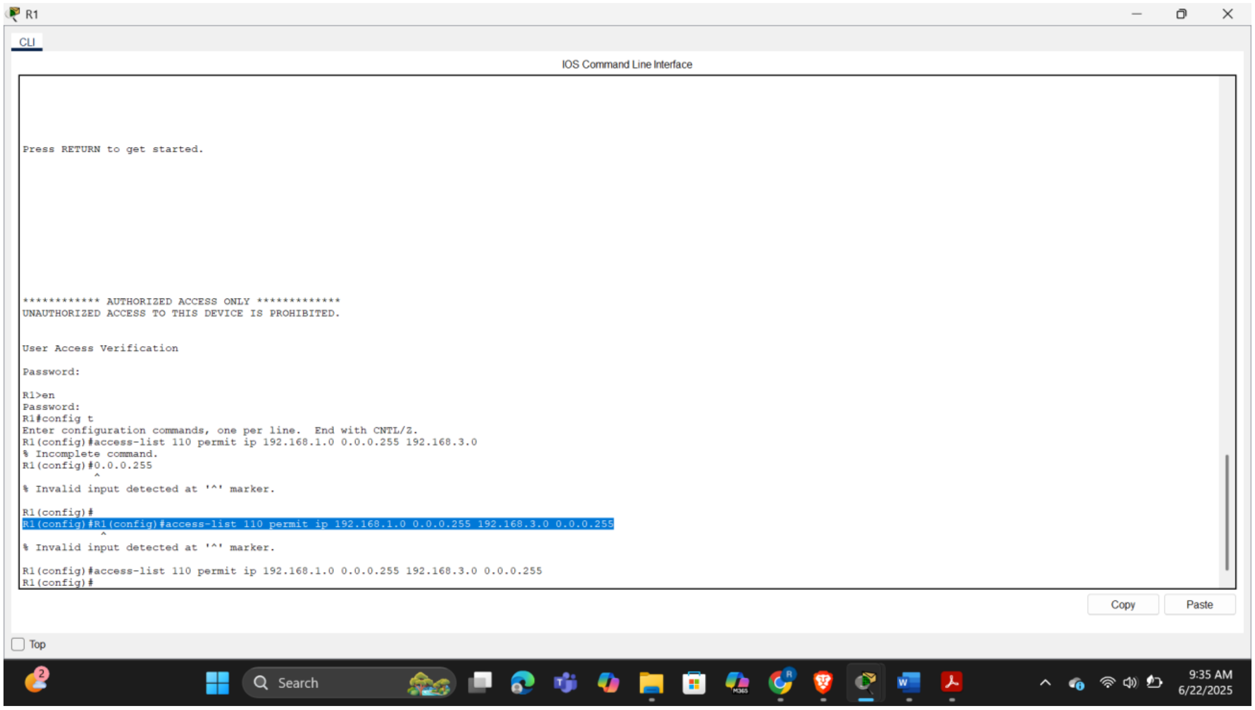

Fig 1.4 identifying interesting traffic

Configuring IKE Phase 1 (ISAKMP Policy)

Next, I set up the parameters for IKE Phase 1, which is responsible for establishing a secure tunnel and negotiating the initial encryption parameters. I created an ISAKMP policy using crypto isakmp policy 10. Inside the policy configuration mode, I specified the encryption method as AES-256 (encryption aes 256), set the authentication method to pre-shared key (authentication pre-share), and selected Diffie-Hellman group 5 (group 5) for key exchange. These settings ensure that the tunnel setup is secure and the keys are exchanged confidentially. After exiting ISAKMP configuration mode, I configured the pre-shared key vpnpa55 to be associated with R3’s public IP (10.2.2.2) using crypto isakmp key vpnpa55 address 10.2.2.2.

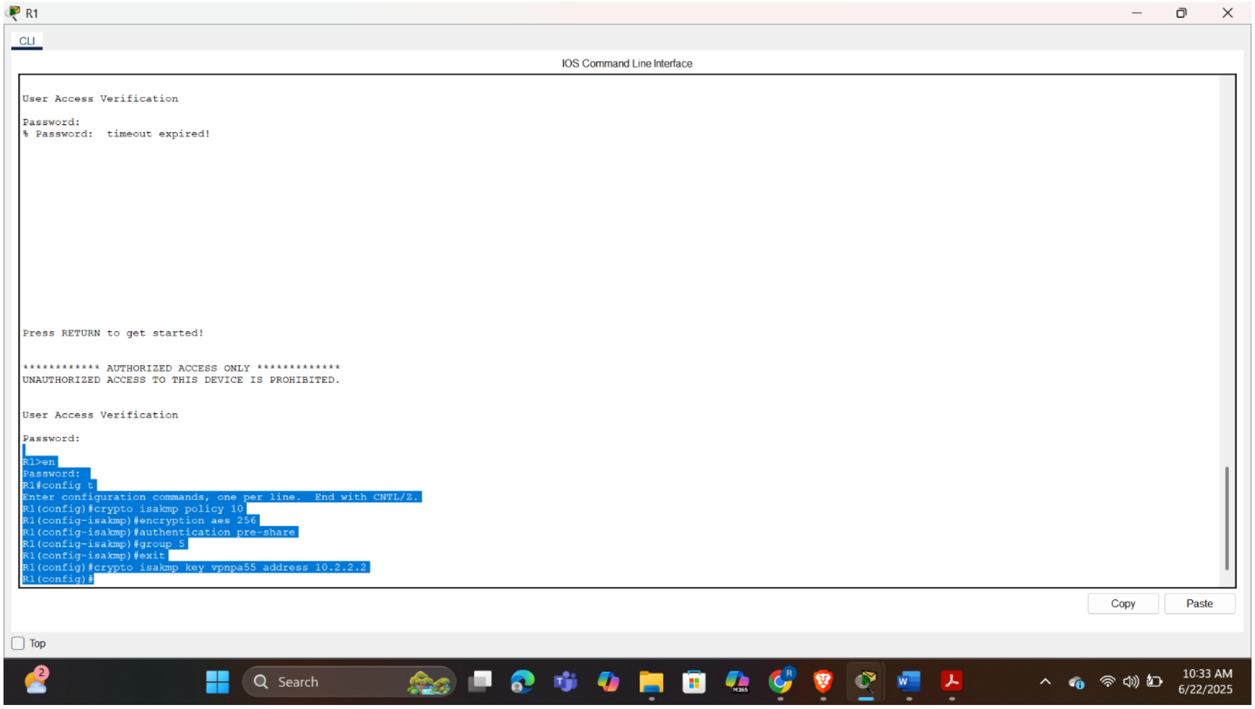

Fig 1.5 Configuring the IKE Phase 1 ISAKMP policy on R1

Configuring IKE Phase 2 (IPsec Policy)

With Phase 1 configured, I moved on to Phase 2, where the actual data encryption settings are defined. I began by creating a transform set named VPN-SET using crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac, which specified that the tunnel should use ESP for both encryption (AES) and authentication (SHA-HMAC). Then, I created a crypto map called VPN-MAP using crypto map VPN-MAP 10 ipsec-isakmp. Inside the crypto map configuration, I added a description to clarify its purpose (description VPN connection to R3), defined the remote peer (set peer 10.2.2.2), linked the transform set (set transform-set VPN-SET), and matched the traffic defined in ACL 110 (match address 110). This crypto map brings together all the components necessary to establish and maintain the secure IPsec tunnel.

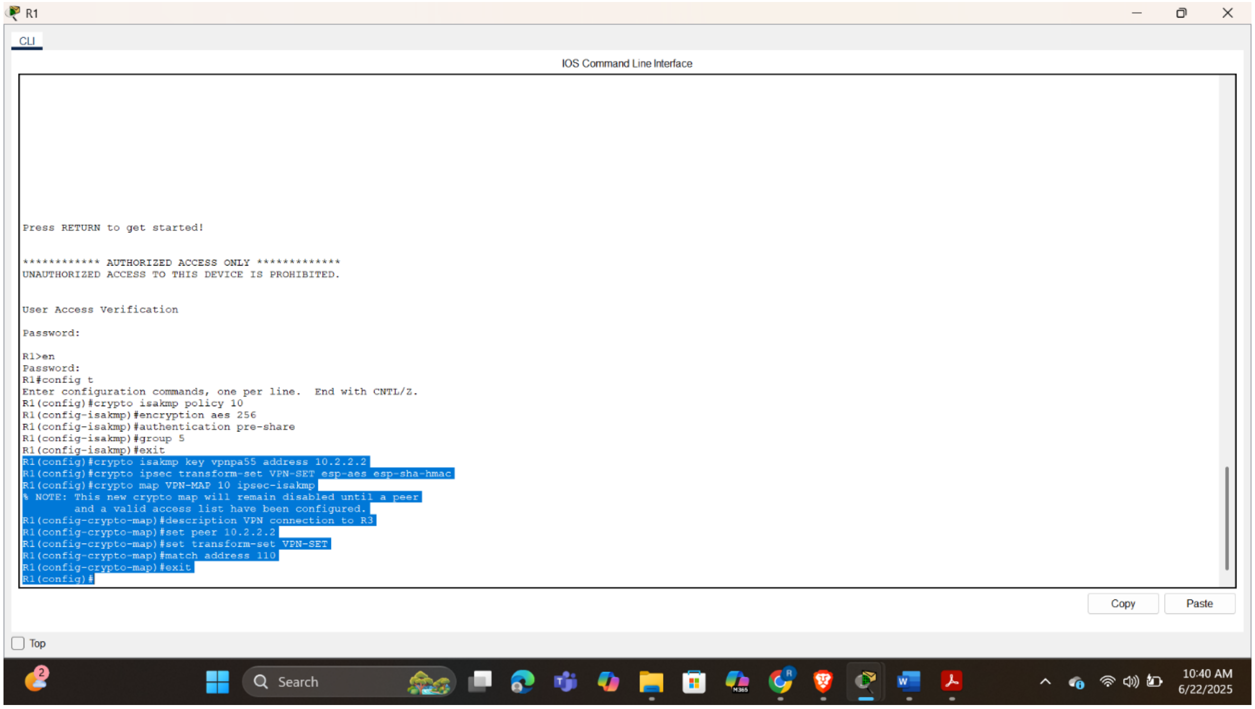

Fig 1.6 Configuring the IKE Phase 2 IPsec policy on R1

Applying the Crypto Map to the Outgoing Interface

Finally, I applied the crypto map to the router's outgoing interface, which is Serial 0/0/0. I entered the interface configuration mode with interface s0/0/0 and then applied the map using crypto map VPN-MAP. This binding ensures that any traffic exiting this interface that matches ACL 110 will trigger the IPsec VPN tunnel to be negotiated and encrypted accordingly.

Fig 1.7 Configuring the crypto map on the outgoing interface

Part 3

Configure IPsec Parameters on R3

To configure the second endpoint of the site-to-site VPN tunnel, I began with R3 by first confirming whether the Security Technology package necessary for VPN support was already enabled. I did this using the show version command. This output revealed whether the securityk9 package was activated. If it had not been active, I would have enabled it using the command license boot module c1900 technology-package securityk9, followed by accepting the end-user agreement, saving the configuration, and reloading the router to activate the security package. Since enabling this package is essential for IPsec functionality, I ensured it was fully operational before proceeding.

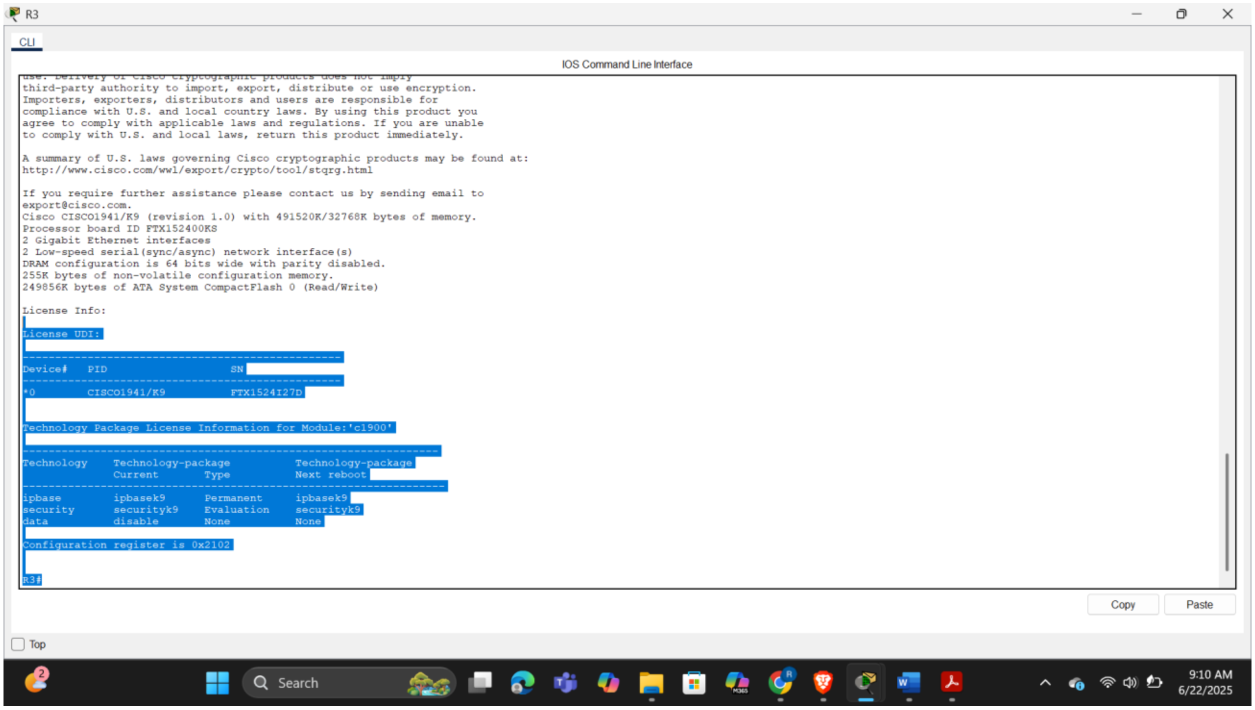

Fig 1.8 verifying security parameters on R3

Configuring router R3 to support a site-to-site VPN with R1

Next, I created an access control list (ACL) on R3 to define interesting traffic that would trigger the VPN tunnel. The ACL was configured using the command access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255, which specifies that only traffic originating from R3's LAN subnet and destined for R1's LAN should initiate the VPN tunnel. This ensures encryption is reserved only for relevant communication, maintaining both performance and security.

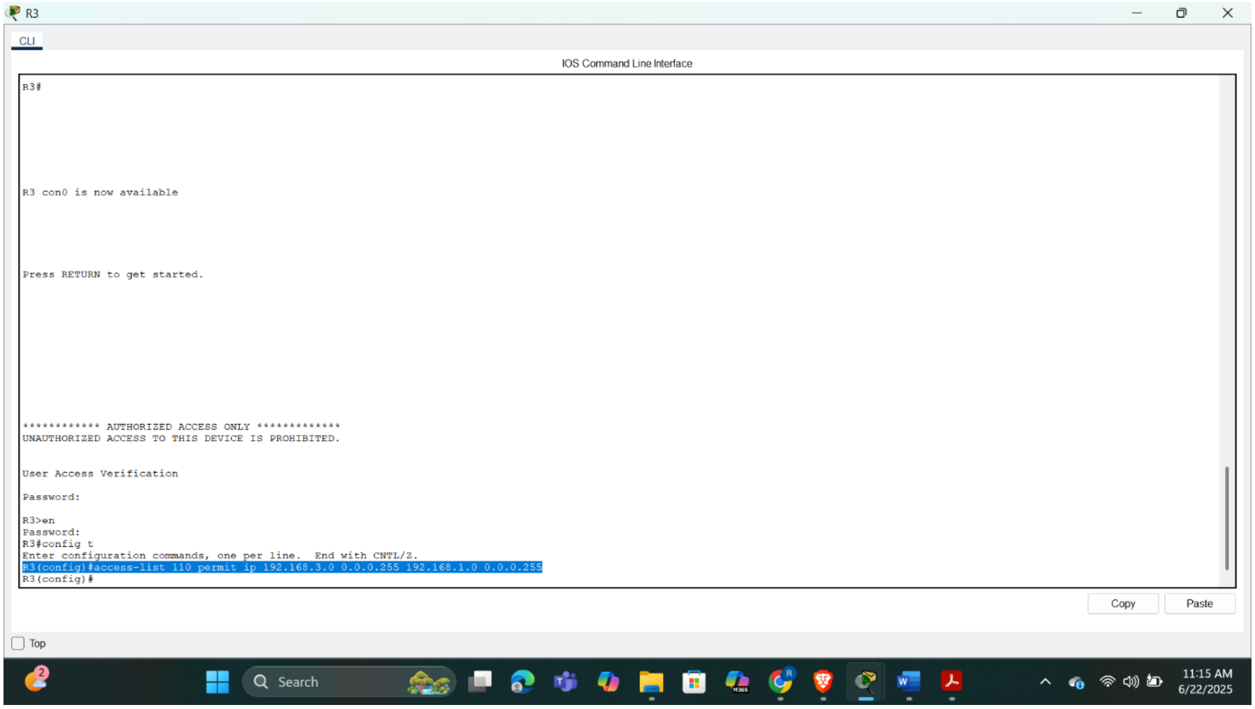

Fig 1.9 Configuring router R3 to support a site-to-site VPN with R1

Configuring the IKE Phase 1 SAKMP properties on R3

With interesting traffic identified, I moved on to setting up the Internet Key Exchange (IKE) Phase 1 policy on R3. I configured policy number 10 with the command crypto isakmp policy 10. Inside this policy, I set the encryption method to AES 256-bit with encryption aes 256, the authentication method to pre-shared keys using authentication pre-share, and specified Diffie-Hellman Group 5 for key exchange via group 5. After exiting the policy configuration, I established the shared key vpnpa55 to be used specifically with R1's WAN interface by entering crypto isakmp key vpnpa55 address 10.1.1.2. This established a secure and consistent key for both ends of the VPN tunnel.

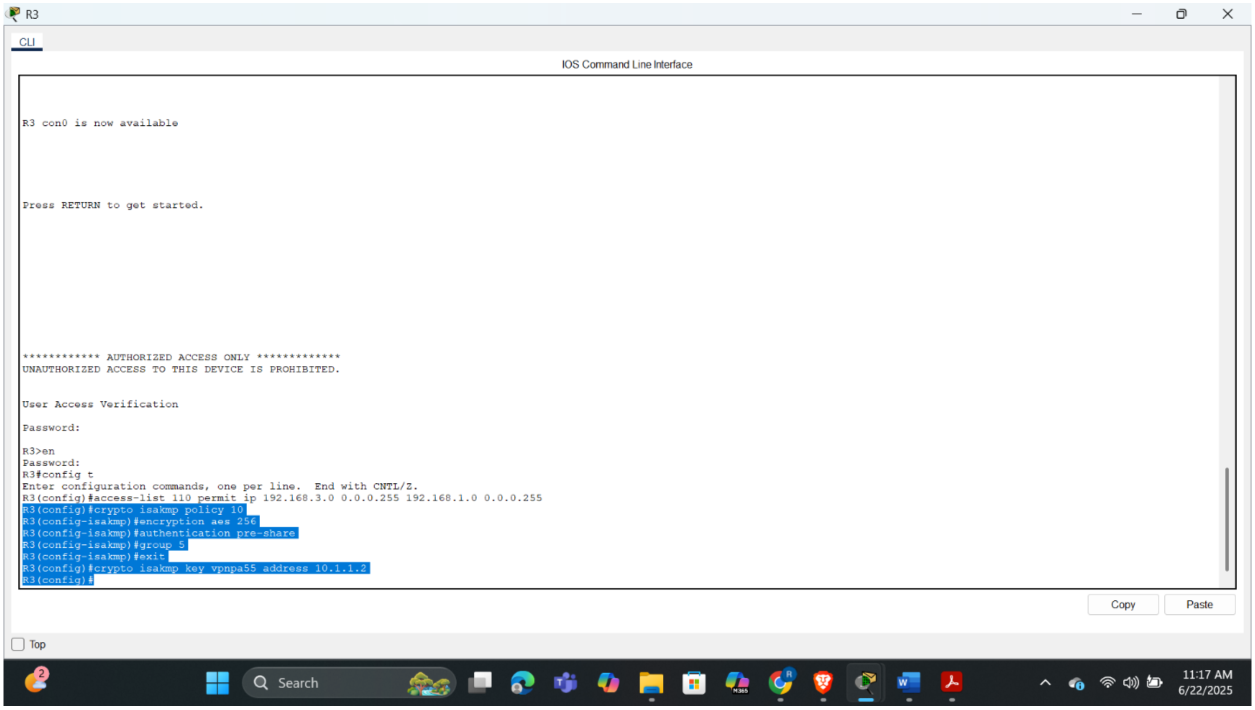

Fig 2.0 Configuring the IKE Phase 1 SAKMP properties on R3.

Configuring the IKE Phase 2 IPsec policy on R3

Following Phase 1 setup, I configured the IKE Phase 2 IPsec policy. I created a transform set named VPN-SET with the command crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac, selecting encryption via AES and hashing via SHA for robust confidentiality and integrity. Then, I created the crypto map named VPN-MAP with sequence number 10 and designated it as an ipsec-isakmp map using crypto map VPN-MAP 10 ipsec-isakmp. Within this map, I added a description for clarity, then set R1's IP address as the peer using set peer 10.1.1.2. I applied the previously defined transform set and matched the crypto map to ACL 110 using match address 110. These settings tie together the parameters necessary for Phase 2 and define how encrypted traffic should be handled.

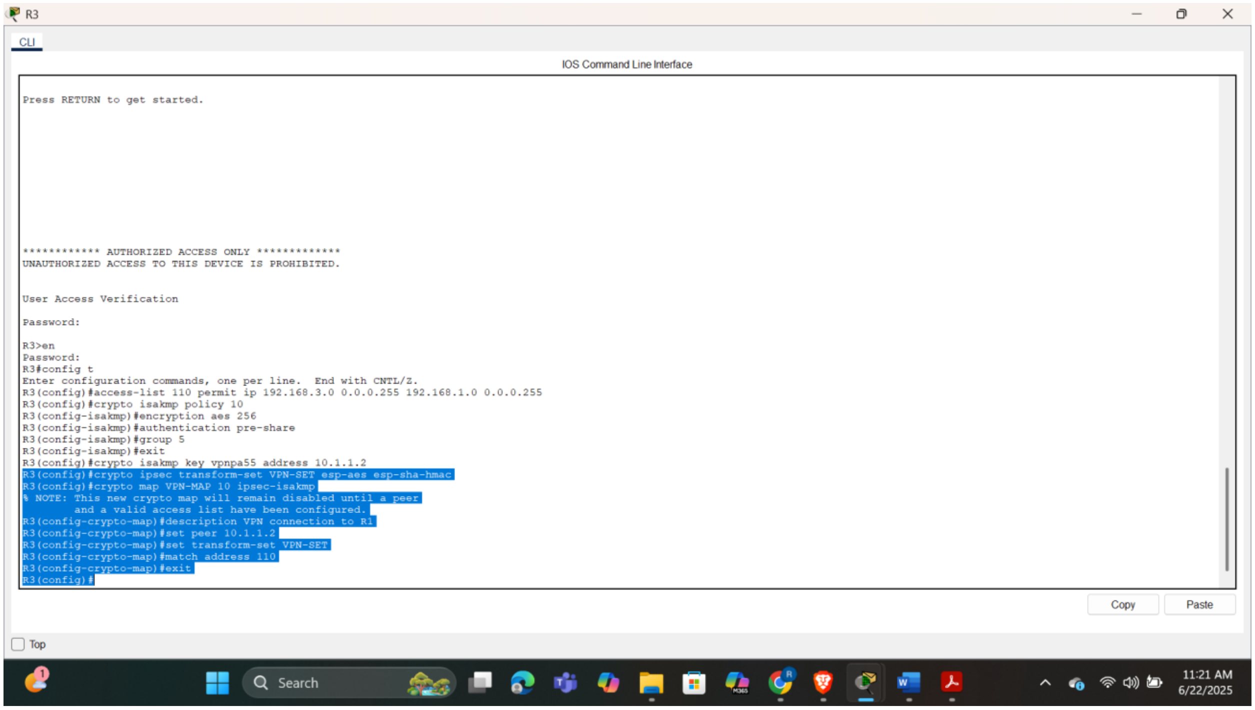

Fig 2.1 Configuring the IKE Phase 2 IPsec policy on R3.

Configuring the crypto map on the outgoing interface

Finally, I applied the crypto map to R3's outgoing interface by navigating to the serial interface connected to R1 using interface s0/0/1, and then I bound the map with the command crypto map VPN-MAP. This step finalized the IPsec VPN configuration on R3 and enabled the router to actively participate in encrypted communication with R1 once interesting traffic is detected

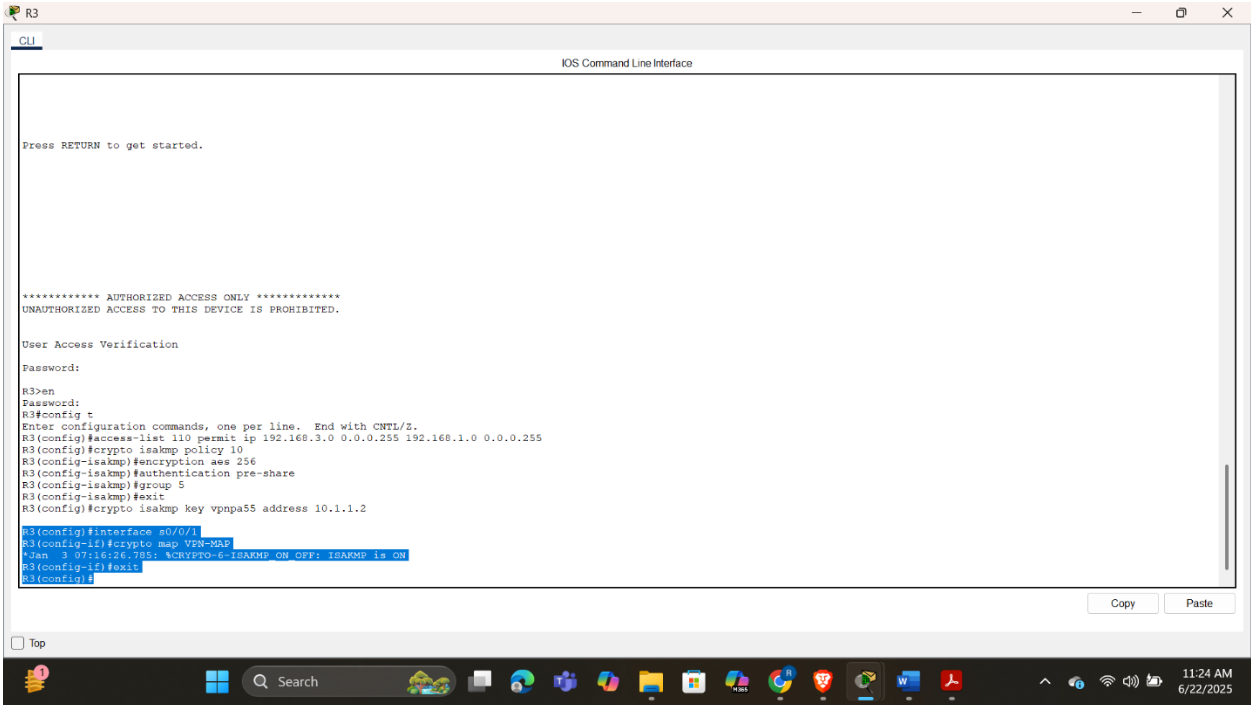

Fig 2.2 Configuring the crypto map on the outgoing interface.

Part 4

Verifying the IPsec VPN

Checking current security statistics

To begin verifying the IPsec tunnel functionality, I first ensured that the tunnel was not active by checking the current security association statistics. On **R1**, I issued the show crypto ipsec sa command prior to initiating any interesting traffic. As expected, all packet counters — including #pkts encaps, #pkts encrypt, #pkts decaps, and #pkts decrypt — were at **0**, confirming that no traffic had yet triggered the tunnel, and it was currently inactive.

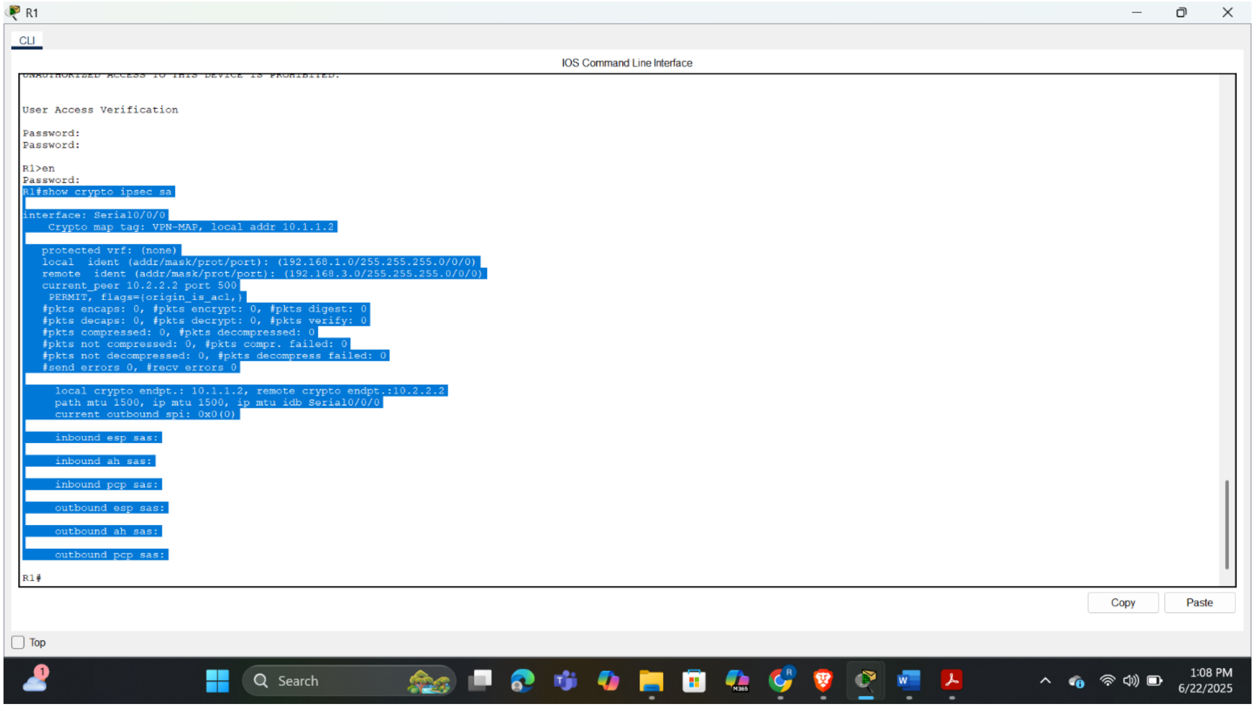

Fig 2.3 Tunnel before interesting traffic

Generating interesting traffic

Next, to generate interesting traffic, I moved to **PC-A** (within R1's LAN at 192.168.1.0/24) and pinged **PC-C** (in R3's LAN at 192.168.3.0/24) using the command ping 192.168.3.3. Since this traffic matched the ACL 110 configuration identifying interesting traffic, it was expected to trigger the creation of the IPsec tunnel. The first ping attempt timed out, which is normal due to the time needed to negotiate ISAKMP and IPsec phases. However, the subsequent pings were successful, indicating the tunnel had been established.

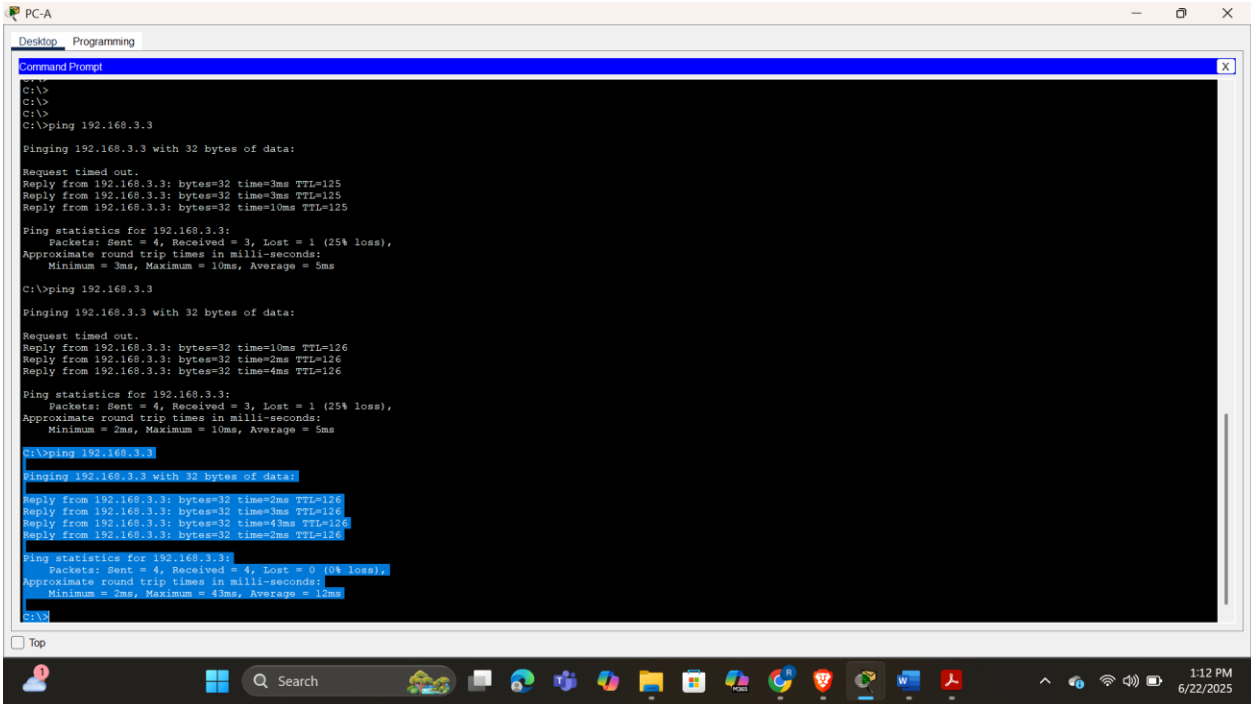

Fig 2.4 Pinging pc C from Pc B

Checking current security statistics after interesting traffic

Returning to **R1**, I re-issued the show crypto ipsec sa command. This time, the output showed non-zero packet counters across the encapsulated, encrypted, decapsulated, and decrypted fields. This confirmed that the IPsec tunnel was now **up and actively processing traffic**, successfully encrypting and decrypting data between the two LANs. The same verification on **R3** yielded matching results, ensuring both ends were synchronized and secure.

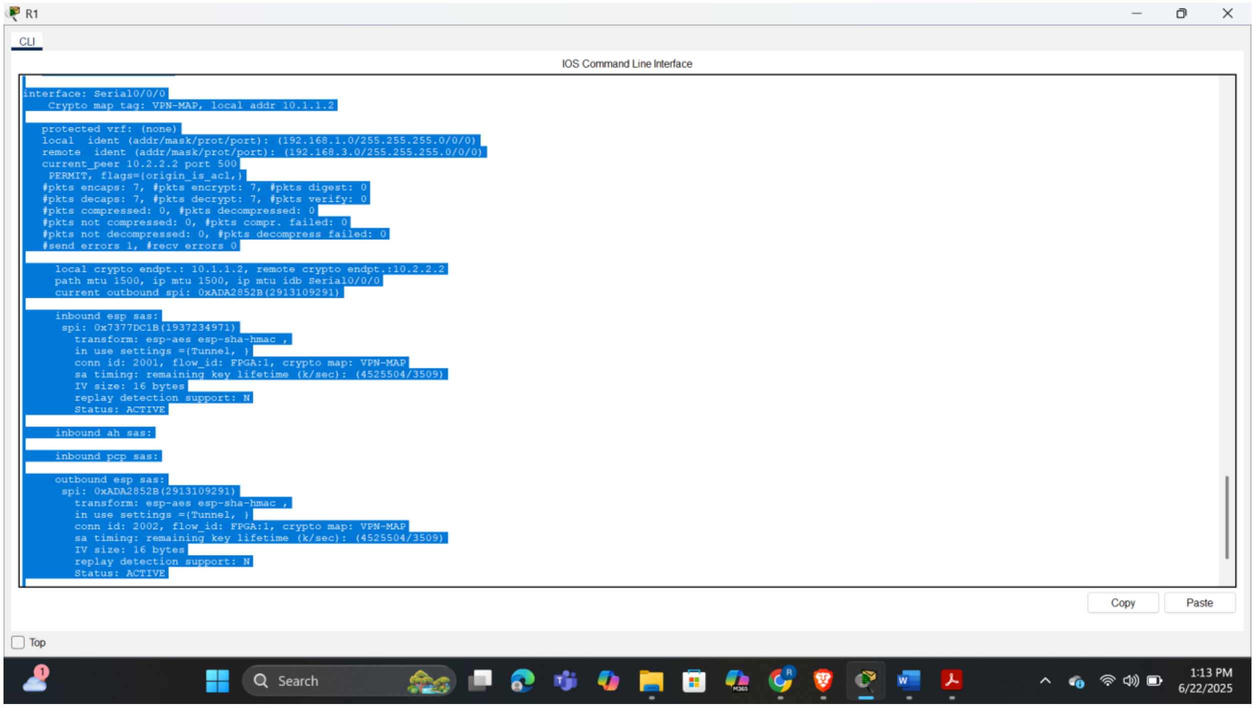

Fig 2.5 Verifying tunnel after interesting traffic

Generating uninteresting traffic

To further validate the VPN's behavior, I then tested how it handles uninteresting traffic. From **PC-A**, I pinged **PC-B** (also on R1's local LAN). Since this traffic does not match the ACL and is local to R1's network, it was not defined as interesting. After the ping completed successfully, I went back to **R1** and re-checked the IPsec statistics with show crypto ipsec sa. As expected, the packet counters had **not changed** since the last interesting traffic ping, confirming that **non-interesting traffic is not being encrypted** and does not engage the VPN tunnel.

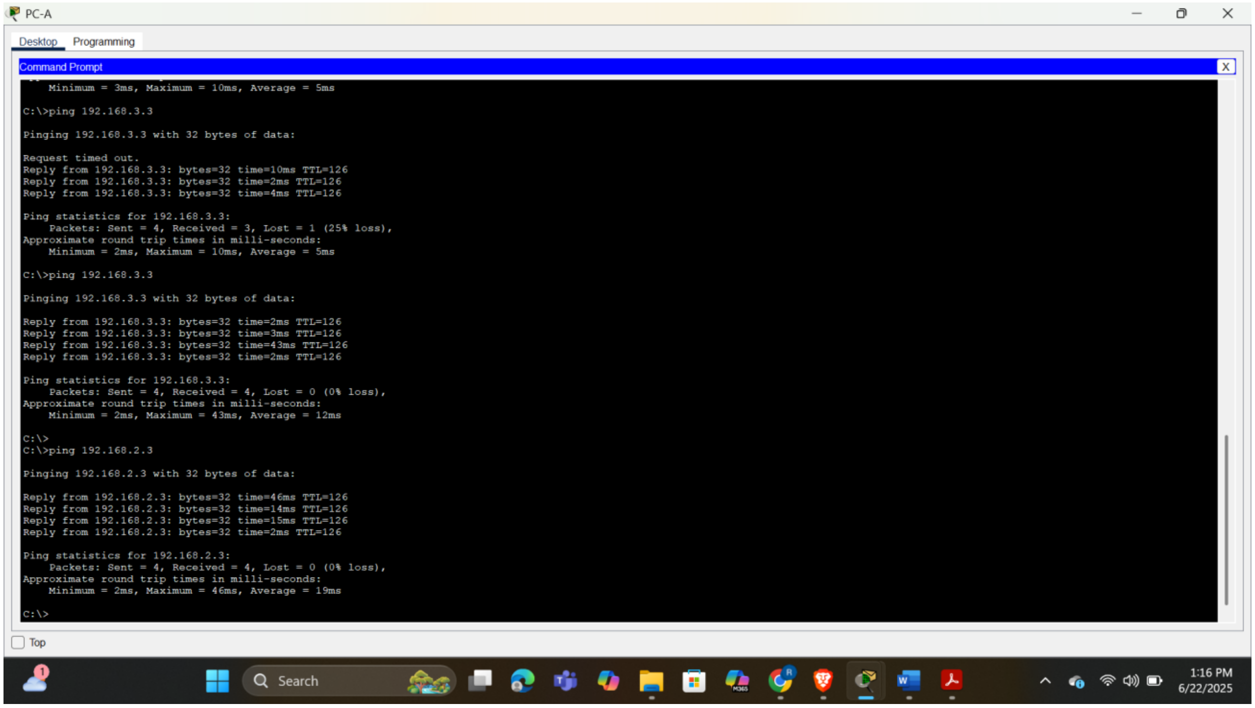

Fig 2.6 Pinging pc B from pc A

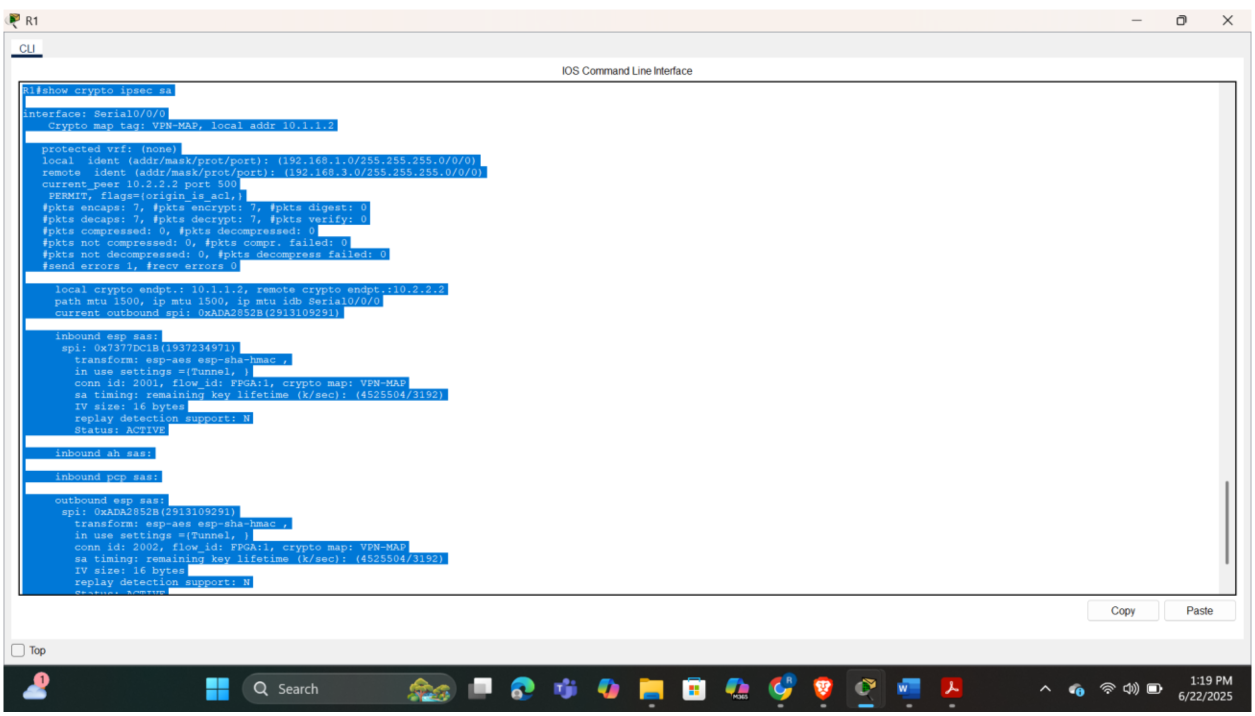

Fig 2.7 tunnel showing No change since there is no interesting traffic

With these observations and command outputs, I concluded that the IPsec Site-to-Site VPN between **R1 and R3** is fully operational. The tunnel is dynamically established only when interesting traffic is detected, encrypts and decrypts packets as expected, and safely ignores non-interesting traffic. This successful end-to-end verification demonstrates that the configuration meets the intended security objectives.

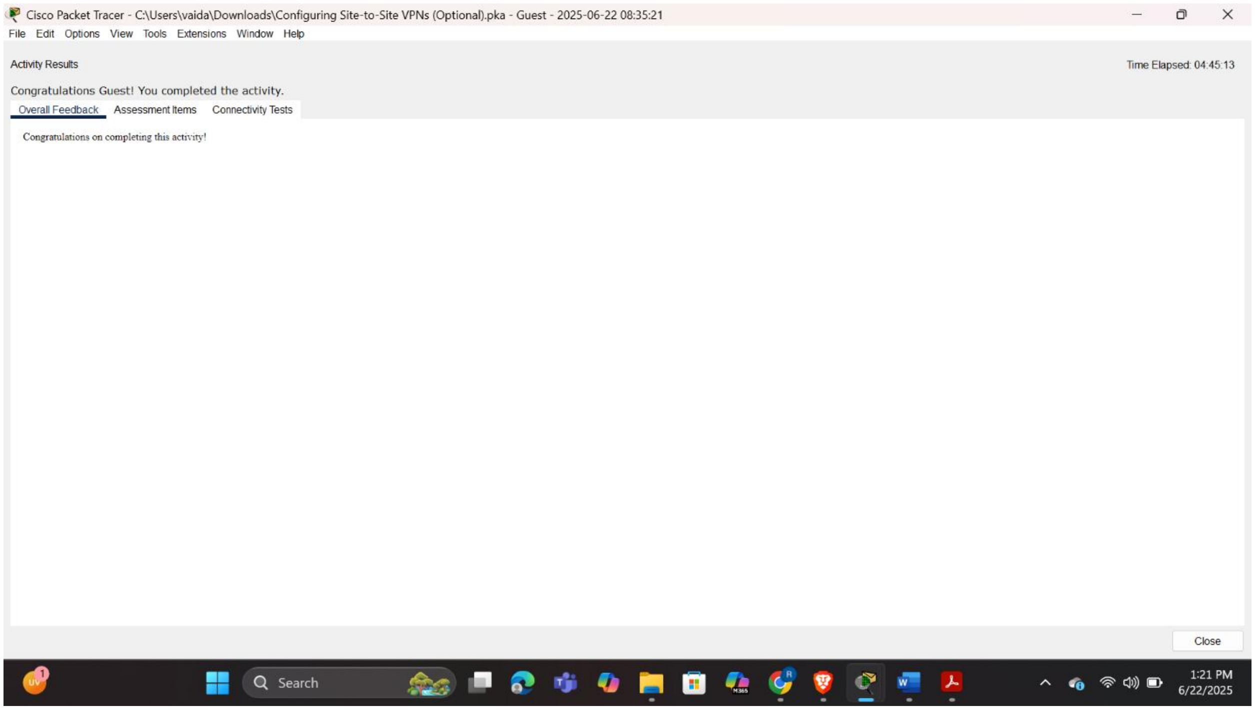

Fig 2.8 completion status

Conclusion

Through this exercise, I successfully configured and tested an IPSec VPN tunnel and gained valuable hands-on experience in interpreting its operational state using Cisco's show crypto ipsec sa command. I observed changes in the tunnel status, such as the progression from inactive to active states triggered by interesting traffic, and monitored the number of encrypted and decrypted packets to confirm proper tunnel functionality. This lab not only reinforced my understanding of IPSec VPN concepts but also enhanced my ability to troubleshoot and validate secure connections in real-time, which is a critical skill in network security administration.